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FA 8

1) An urn contains 10 red, 8 black, and 7 green balls. One of the colors is chosen at random (meaning that the chosen color is equally likely to be any of the 3 colors), and then 4 balls are randomly chosen from the urn.
Let X be the number of those balls that are of the chosen color.

(a) Find $P(X=0)$

$N = 10+8+7 = 25$

if red is chosen:

$$P(X=0|R) = \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{1365}{12650} = \frac{273}{2530}$$

if green is chosen:

$$P(X=0|G) = \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{3060}{12650} = \frac{306}{1265}$$

if black is chosen:

$$P(X=0|B) = \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{2380}{12650} = \frac{238}{1265}$$

$$P(X=0) = \frac{1}{3}\left[P(X=0|R) + P(X=0|B) + P(X=0|G)\right]$$

$$= \frac{1}{3}\left[\frac{1365}{12650} + \frac{2380}{12650} + \frac{3060}{12650}\right] = \frac{1}{3}\left(\frac{6805}{12650}\right) = \frac{6805}{37950}$$

$$\boxed{P(X=0) = 0.1793}$$

(b) Let X_i equal 1 if the ith ball selected is of the chosen color, and let it equal 0 otherwise. Find $P(X_i=1)$, $i=1,2,3,4$

$$P(X_i=1) = \sum_{C \in \{R,B,G\}} P(X_i=1|C)\cdot P(C)$$

$$\left.\begin{aligned} P(X_i=1|R) &= \frac{10}{25} \\ P(X_i=1|B) &= \frac{8}{25} \\ P(X_i=1|G) &= \frac{7}{25} \end{aligned}\right\} \quad P(X_i=1) = \frac{1}{3}\left(\frac{10}{25} + \frac{8}{25} + \frac{7}{25}\right) = \frac{1}{3} \approx 0.3333$$

$$\boxed{P(X_i=1) = 0.3333 \quad \text{for } i=1,2,3,4}$$

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(c) Find $E[X]$

Hint: Express X in terms of X_1, X_2, X_3, and X_4

$$X = X_1 + X_2 + X_3 + X_4$$

$$E[X] = E[X_1 + X_2 + X_3 + X_4] = E[X_1] + E[X_2] + E[X_3] + E[X_4]$$

since x_i is an indicator random variable

$$E[X_i] = P(X_i = 1) = \frac{1}{3}$$

$$\boxed{E[X] = 4 \times \frac{1}{3} = \frac{4}{3} \approx 1.3333}$$

$$\boxed{E[X] = 1.3333}$$

2) The random variable X is said to have the Yule-Simon distribution if

$$P(X = n) = \frac{4}{n(n+1)(n+2)} \qquad n \geq 1$$

(a) Show that the preceding is actually a probability mass function. That is, show that $\sum_{n=1}^{\infty} P(X = n) = 1$

$$4\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right] = 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$= 2\left[\left(\frac{1}{1(2)} - \frac{1}{2(3)}\right) + \left(\frac{1}{2(3)} - \frac{1}{3(4)}\right) + \left(\frac{1}{3(4)} - \frac{1}{4(5)}\right) + \cdots\right]$$

$$= 2\left(\frac{1}{2}\right) = 1$$

$$\boxed{\sum_{n=1}^{\infty} P(X = n) = 1}$$

(b) Show that $E[X] = 2$

$$E[X] = \sum_{n=1}^{\infty} n\,P(X = n) = \sum_{n=1}^{\infty} n\left(\frac{4}{n(n+1)(n+2)}\right) = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

$$\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2} \quad \text{so,} \quad 4\sum_{n=1}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = 4\left[\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \cdots\right] = 4\left(\frac{1}{2}\right) = 2$$

$$\boxed{E[X] = 2 //}$$

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(c) Show that

$$E[X^2] = \infty$$

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \cdot P(X=n) = \sum_{n=1}^{\infty} n^2 \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}$$

Analyzing the behavior of the terms as $n \to \infty$:

$$\frac{n}{(n+1)(n+2)} = \frac{n}{n^2+3n+2} \sim \frac{1}{n}$$

LIMIT COMPARISON TEST

$$\lim_{n\to\infty} \frac{\frac{n}{(n+1)(n+2)}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n^2+3n+2} = 1 > 0$$

since $1/n$ diverges, $\frac{n}{(n+1)(n+2)}$ also diverges

$$\therefore \boxed{E[X^2] = \infty}$$

3) The probability density function of X is given by

$$f(x) = \begin{cases} a+bx^2, & 0 \le x < 1, \\ 0, & \text{otherwise,} \end{cases}$$

if $E[X] = \frac{3}{5}$, find the values of a and b:

$$\int_0^1 f(x)\,dx = 1 \to \int_0^1 (a+bx^2)\,dx = \left[ax + \frac{bx^3}{3}\right]_0^1 = a + \frac{b}{3} = 1$$

$$\to 3a + b = 3$$

$$E[X] = \int_0^1 x f(x)\,dx = \int_0^1 (ax+bx^3)\,dx = \left[\frac{ax^2}{2} + \frac{bx^4}{4}\right]_0^1 = \left[\frac{a}{2} + \frac{b}{4} = \frac{3}{5}\right] \cdot 20$$

$$\Rightarrow 10a + 5b = 12$$

$b = 3 - 3a$,

$10a + 5(3-3a) = 12$

$10a + 15 - 15a = 12$

$a = \frac{3}{5}$

$b = 3 - 3a$

$b = 3 - 3\left(\frac{3}{5}\right)$

$b = \frac{6}{5}$

$$\boxed{a = \frac{3}{5},\ b = \frac{6}{5}}$$

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4) Let X be a random variable having expected value μ and variance σ^2. Find the expected value and variance of

$$Y = \frac{X-\mu}{\sigma}$$

$E[aX+b] = aE[X]+b$ $\qquad$ $E[Y] = \frac{1}{\sigma}E[X-\mu]$

$$E[Y] = \frac{1}{\sigma}(E[X]-\mu)$$

Substitute $E[X]=\mu$:

$$E[Y] = \frac{1}{\sigma}(\mu-\mu) = 0$$

Find the variance:

$Var(aX+b) = a^2 Var[X]$ $\qquad$ $Y = \frac{X-\mu}{\sigma}$

$$Y = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$$

Therefore,

$$Var(Y) = Var\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right)$$

$$Var(Y) = \left(\frac{1}{\sigma}\right)^2 Var(X)$$

$$Var(Y) = \left(\frac{1}{\sigma^2}\right) Var(X)$$

→ Substitute $Var(X) = \sigma^2$:

$$\therefore Var(Y) = \frac{1}{\sigma^2}\cdot\sigma^2 = 1$$

$$\boxed{E[Y] = 0.0000 \text{ and } Var(Y) = 1.0000}$$

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5) Let X and Y be independent ~~variable~~ Poisson random variables with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$, respectively. Let $W = X + Y$

(a) Using property of ~~values~~ sums of independent random variables, derive the moment generating function (MGF) $M_W(t)$ of W.

Let $X \sim \text{Poisson}(\lambda_1 = 2)$ and $Y \sim \text{Poisson}(\lambda = 3)$ be independent random variables, and $W = X + Y$

$$M(t) = e^{\lambda(e^t - 1)}$$

thus, $M_X(t) = e^{2(e^t-1)}$

$M_Y(t) = e^{3(e^t-1)}$

since X and Y are independent:

$$M_W(t) = E[e^{tW}] = E[e^{t(X+Y)}] = E[e^{tX}e^{tY}] = E[e^{tX}]E[e^{tY}] = M_X(t) \cdot M_Y(t)$$

$$M_W(t) = e^{2(e^t-1)} \cdot e^{3(e^t-1)} = e^{(2+3)(e^t-1)} = e^{5(e^t-1)}$$

$$\boxed{M_W(t) = e^{5(e^t-1)}}$$

(b) Based on the uniqueness property of MGF's, identify the probability distribution and parameter(s) of W.

$$M_W(t) = e^{5(e^t-1)} \quad M(t) = e^{\lambda(e^t-1)}$$

$$\lambda = 5$$

since the MGF uniquely identifies a probability distribution, W follows a Poisson distribution

$$\boxed{\begin{array}{c} W \sim \text{Poisson}(5) \\ \text{Parameter: } \lambda_W = \lambda_1 + \lambda_2 = 2 + 3 = 5 \end{array}}$$

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(c) Determine the expected value $E[W]$ and the second moment $E[W^2]$ by successively differentiating $M_W(t)$ and evaluating it at $t=0$

$$M_W(t) = e^{5(e^t-1)}$$

MGF moment formulas:

$E[W] = M'_W(0)$

$E[W^2] = M''_W(0)$

$u = 5(e^t-1)$

$u' = 5e^t$

$\frac{d}{dt} e^u = e^u \cdot u'$

$M'_W(t) = 5e^t e^{5(e^t-1)}$

$E[W] = 5e^0 e^{5(e^0-1)}$

$$\boxed{E[W] = 5.0000}$$

$M'_W(t) = 5e^t e^{5(e^t-1)}$

Let $u = 5e^t$ $\quad u' = 5e^t$

$v = e^{5(e^t-1)}$ $\quad u' = 5e^t e^{5(e^t-1)}$

$M''_W(t) = 5e^t e^{5(e^t-1)} + 5e^t\left(5e^t e^{5(e^t-1)}\right)$

$= \left(5e^t + 25e^{2t}\right)\left(e^{5(e^t-1)}\right)$

$E[W^2] = M''_W(0)$

$E[W^2] = \left(5e^0 + 25e^{2(0)}\right)\left(e^{5(e^0-1)}\right)$

$= (5+25)\, e^{5(0)}$

$= 30$

$$\boxed{E[W^2] = 30.0000}$$

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